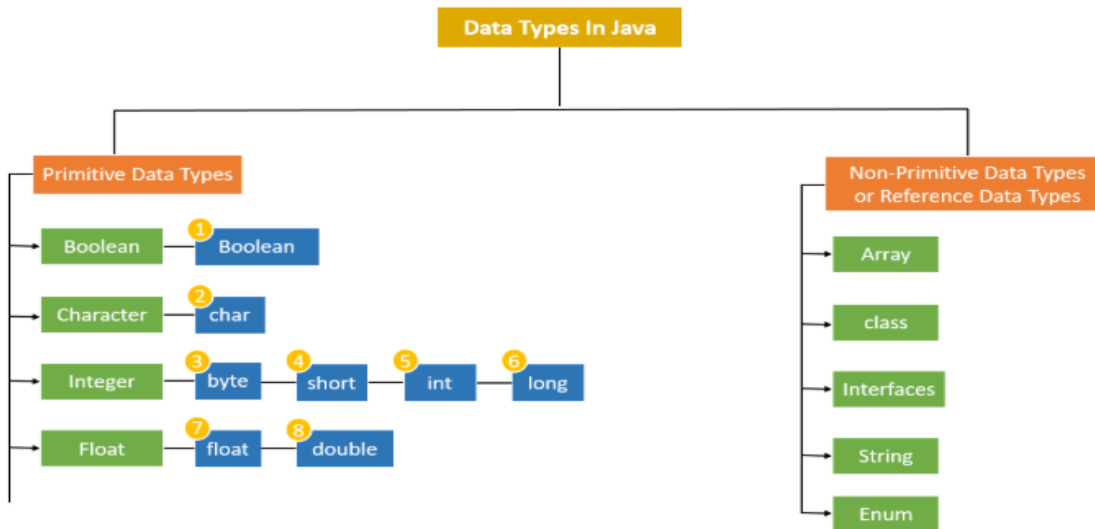


Primitive Data types

Data types in Java specify the size and type of values that a variable can store. Java is a **strongly typed language**, meaning every variable must be declared with a specific data type before it can be used. Java data types are divided into two main categories: **Primitive** and **Non-Primitive**.



TYPE	DESCRIPTION	DEFAULT	SIZE	EXAMPLE LITERALS	RANGE OF VALUES
boolean	true or false	false	1 bit	true, false	true, false
byte	twos complement integer	0	8 bits	(none)	-128 to 127
char	unicode character	\u0000	16 bits	'a', '\u0041', '\101', '\l', '\', '\n', '\b'	character representation of ASCII values 0 to 255
short	twos complement integer	0	16 bits	(none)	-32,768 to 32,767
int	twos complement integer	0	32 bits	-2, -1, 0, 1, 2	-2,147,483,648 to 2,147,483,647
long	twos complement integer	0	64 bits	-2L, -1L, 0L, 1L, 2L	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
float	IEEE 754 floating point	0.0	32 bits	1.23e100f, -1.23e-100f, .3f, 3.14F	upto 7 decimal digits
double	IEEE 754 floating point	0.0	64 bits	1.23456e300d, -1.23456e-300d, 1e1d	upto 16 decimal digits

Numeric Types (Integers): Used to store whole numbers without decimals.

Data Type	Size	Memory Range	Common Use Case
<code>byte</code>	1 byte (8 bits)	-128 to 127	Saving memory in large arrays
<code>short</code>	2 bytes (16 bits)	-32,768 to 32,767	Rare; used for memory constraints
<code>int</code>	4 bytes (32 bits)	-2.14 Billion to 2.14 Billion	Default choice for whole numbers
<code>long</code>	8 bytes (64 bits)	-9 Quintillion to 9 Quintillion	Large numbers (IDs, system time)

Note: When assigning a value to a long, append an L or l to the end of the number (e.g., long population = 8000000000L;)

Numeric Types (Floating-Point): Used to store fractional numbers containing decimals.

Data Type	Size	Precision	Common Use Case
<code>float</code>	4 bytes (32 bits)	6 to 7 decimal digits	Game graphics, saving memory
<code>double</code>	8 bytes (64 bits)	Up to 15 decimal digits	Default choice for math and decimals

⚠ **Note:** Java treats all decimal numbers as `double` by default. To declare a `float`, you must append an `F` or `f` to the value (e.g., `float price = 19.99f;`).

Character Type: Used to store a single character.

Data Type	Size	Contents	Description
<code>char</code>	2 bytes (16 bits)	Single character or ASCII	Uses single quotes (<code>'A'</code> , <code>'3'</code> , <code>'\$'</code>). Uses Unicode system.

Boolean Type: Used to store conditional states.

Data Type	Size	Allowed Values	Common Use Case
<code>boolean</code>	1 bit (virtual)	<code>true</code> or <code>false</code>	Conditional flags (<code>if/else</code> logic)